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Theoretical Physics

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Introduction

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1.1 | Symmetry Groups in Quantum Physics

Symmetry transformations are operations carried out on the devices which prepare the states and the apparatuses which measure the observables of a quantum system having the property of not changing the results of experiments. For instance, space translations are symmetry transformations since the uniform translation of all preparations devices and measuring apparatuses does not change the results of experiments due to the homogeneity of space. Likewise, space rotations are symmetry transformations since the uniform rotation of all preparations devices and measuring apparatuses does not change the results of experiments due to the isotropy of space. In relativity, Lorentz boosts are symmetry transformations since as the result of experiments are the same in a given inertial frame and any inertial frame moving at uniform speed with respect to the first because the isotropy of spacetime.

In general, symmetry transformations gather together in families with special properties. Let G be one such family. Carrying out successively two symmetry transformations g, g' of G yields a third symmetry transformation $g' \cdot g$ of G called the **product** of g, g'. The effect of a symmetry transformation g of G can be undone by another symmetry transformation g^{-1} of G called the **inverse** of g. There exists a distinguished symmetry transformation e of G that has no effect called **neutral**. The multiplication and inversion operations and neutral element of G have the following properties:

- **Associativity**: for any three symmetry transformations g, g', g'' of G we have
$$(g'' \cdot g') \cdot g = g'' \cdot (g' \cdot g). \tag{1.1.1}$$
- **Neutrality**: there exists a symmetry transformation e of G such that for any symmetry transformation g of G we have
$$e \cdot g = g \cdot e = g. \tag{1.1.2}$$
- **Invertibility**: for any symmetry transformation g of G there exists a symmetry transformation g^{-1} of G such that
$$g^{-1} \cdot g = g \cdot g^{-1} = e. \tag{1.1.3}$$

These relations characterize G as an algebraic structure called a **group**. Since G is constituted by symmetry transformations, G is called more specifically a **symmetry group**. A group G is called **finite** if it is constituted by a finite number of elements, else it is called **infinite**. A group G is called **Abelian** if multiplication turns out to be commutative, that is

$$g \cdot g' = g' \cdot g, \quad \forall g, g' \in G. \tag{1.1.4}$$

Else, it is called non Abelian. The groups encountered in quantum physics are bot finite and infinite and Abelian and non Abelian.

TODO: Complete the example

Example (Space Translation Symmetry). [...]

TODO: Complete the example

Example (Space Rotation Symmetry). [...]

1.1.1 | Symmetry Groups and Their Operator Realization

In quantum theory, states and observables of a system are represented respectively by normalized vectors defined up to a multiplicative phase in a Hilbert space $\mathcal{H}$ and selfadjoint operators in the operator algebra $L(\mathcal{H})$. The invariance of experimental results under the action of a symmetry group reduces to the invariance of expectation values: if Ψ is a state, O is an observable and g is a symmetry transformation of G, ${}^g\Psi$ is the transformed state of Ψ and ${}^gO$ is the transformed observable of O, one must have

$$\langle {}^g\Psi | {}^g O \, | {}^g\Psi\rangle = \langle \Psi | \, O \, | \Psi \rangle \, . \tag{1.1.5}$$

Let us now analyze the implications of the invariance of expectation values under the action of a symmetry group G with a simple example.

Example (“yes–no” observables). As is well–known, there is a class of elementary observables, the “yes–no” observables, defined by taking the value 1, 0 according to whether the system is in a certain state Φ or not. The associated operator is given by

$$O_\Phi \Psi = \Phi \, \langle \Phi | \Psi \rangle \, .$$

It is natural to assume that under the symmetry transformation g of G the observable O_Φ transforms accordingly with the state Φ as follows:

$${}^gO_\Phi = O_{{}^g\Phi}.$$

The expectation value of O_Φ in the state Ψ' is given by the transition probability from Ψ' to Φ:

$$\langle \Psi' | \, O_\Phi \, | \Psi' \rangle = | \, \langle \Phi | \Psi' \rangle \, |^2,$$

such that the **invariance of expectation values** under the action of the symmetry group G implies that

$$| \, \langle {}^g\Phi | {}^g\Psi' \rangle \, |^2 = | \, \langle \Phi | \Psi' \rangle \, |^2.$$

This means that the symmetry transformation g of G must be implemented by a unitary or antiunitary operator U_g on the Hilbert space $\mathcal{H}$ of the system such that

$$| {}^g\Psi \rangle = U_g \, | \Psi \rangle \, , \quad {}^gO = U_g O U_g^{-1}.$$

In this example we have highlighted the importance of the **Wigner's theorem**:

Theorem 1.1 (Wigner's theorem)**.** *Let $\mathcal{H}$ be a Hilbert space and let G be a symmetry group of a quantum system represented on $\mathcal{H}$. Then, for each symmetry transformation g of G, there exists a unitary or antiunitary operator U_g on $\mathcal{H}$ such that*

$$\begin{cases} {}^g\Psi = U(g)\Psi, \\ {}^gO = U_g O U_g^\dagger, \end{cases} \tag{1.1.6}$$

since U_g is unitary or antiunitary, $U_g^{-1} = U_g^\dagger$, and so we can write the transformation of observables as $O \mapsto U_g O U_g^\dagger$. This implies that the expectation value of any observable O in any state Ψ is invariant under the action of the symmetry group G.

This is a fundamental result in quantum theory since it allows us to represent symmetry transformations as unitary or antiunitary operators on the Hilbert space of the system.

Recall that a unitary operator U on a Hilbert space $\mathcal{H}$ is a linear operator that preserves the Hilbert inner product,

$$U(\Phi\, a,\, \Phi'\, a') = U(\Phi)a,\, U(\Phi')a'$$
$$\langle U\Phi | U\Phi'\rangle = \langle \Phi|\Phi'\rangle\,,$$

while an antiunitary operator A on a Hilbert space $\mathcal{H}$ is an antilinear operator that preserves the Hilbert inner product,

$$A(\Phi\, a,\, \Phi'\, a') = A(\Phi)a^*,\, A(\Phi')a'^*$$
$$\langle A\Phi | A\Phi'\rangle = \langle \Phi|\Phi'\rangle^*\,.$$

in both cases the adjoint operator $U^\dagger$ (or $A^\dagger$) of a unitary (or antiunitary) operator U (or A) are equal to the inverse operator U^{-1} (or A^{-1}):

$$\begin{cases} \langle \Psi'|U\Psi\rangle = \left\langle U^\dagger\Psi'\middle|\Psi\right\rangle, \\ \langle \Psi'|A\Psi\rangle = (\left\langle A^\dagger\Psi'\middle|\Psi\right\rangle)^*, \end{cases} \qquad \Longrightarrow \qquad \begin{cases} U^{-1} = U^\dagger, \\ A^{-1} = A^\dagger, \end{cases}$$

respectively for the linear and antilinear case. In what follows, we leave aside the antiunitary case, which occurs only for symmetry transformations involving time reversal, and concentrate on the unitary one. Letting the symmetry transformation g vary in the group G, we obtain a family of unitary operators $U(g)$ contained in the operator algebra $L(\mathcal{H})$.

Let us come back to relation $(1.1.6)_1$. We recall again that state vectors are normalized vectors defined up to a multiplicative phase factor. The content of $(1.1.6)_1$ is that phase choice can be adjusted so that for any symmetry transformation g the state correspondence $\Psi \mapsto^g \Psi$ is codified by a unitary operator $U(g)$. This however does not completely fix $U(g)$: we are still free to replace $U(g)$ by

$$\hat{U}(g) = \alpha(g)U(g), \tag{1.1.7}$$

where $\alpha(g)$ is a complex number of unit modulus, which is called a **phase factor**. This involves a phase redefinition

$${}^g\Psi \to \alpha(g)\, {}^g\Psi,$$

in $(1.1.6)_1$, which does not change the physical content of the transformation since states are defined up to a multiplicative phase factor. By a suitable choice of the phase factors $\alpha(g)$ many relations simplyfy.

It is natural and possible to require

$${}^e\Psi = \Psi, \quad \forall \Psi \in \mathcal{H},$$

so that the neutral element e of the symmetry group G is represented by the identity operator

$$U(e) = \mathbb{I},$$

on the Hilbert space $\mathcal{H}$ of the system. With this choice, we can now analyze the implications of the group properties of G on the family of unitary operators $U(g)$. Considering two symmetry transformations g, g' of G, and a physical state Ψ, we have ${}^{g'}({}^{g}\Psi)$ and ${}^{g'\cdot g}\Psi$ defined only up to a conventional choice of a multiplicative **phase factor** as seen above, one cannot expect that a relation of the form ${}^{g'}({}^{g}\Psi) ={}^{g'\cdot g} \Psi$ to hold strictly in general, but only up to some phase factor

$$\gamma(g, g', \Psi) \in \mathbb{C}, \quad |\gamma(g, g', \Psi)| = 1,$$

depending on the symmetry transformations g, g' and the state Ψ. Thus we can write

$${}^{g'}({}^{g}\Psi) ={}^{g'\cdot g} \Psi\, \gamma(g, g', \Psi).$$

It can be proven that the phase factor $\gamma(g, g', \Psi)$ can be chosen to be independent of the state Ψ, thus obtaining a relation of the form

$$U(g \cdot g') = \gamma(g, g')U(g)U(g'), \tag{1.1.8}$$

which is called a **projective representation** of the symmetry group G. If the phase factor $\gamma(g, g')$ can be chosen to be 1 for all g, g' in G, then we have a **unitary representation** of the symmetry group G.

The phase factors $\gamma(g, g')$ appearing in the projective representation of the symmetry group G are not arbitrary, but must satisfy some consistency conditions, for example the *2-cocyclicity*: given three symmetry transformations g, g', g'' of G, the following relation must hold:

$$\gamma(g, g' \cdot g'')\gamma(g', g'') = \gamma(g \cdot g', g'')\gamma(g, g'), \tag{1.1.9}$$

where functions $\gamma : G\times G \to U(1)$ satisfying the 2-cocyclicity condition are called **2-cocycles** of G. The 2-cocycle condition is a consequence of the associativity property of the group multiplication of G.

The G 2-cocycle $\gamma(g, g')$ enjoys a few special properties, which albeit technical are often useful at various points:

$$\begin{aligned} \gamma(e, g) = \gamma(g, e) = 1, \\ \gamma(g, g^{-1}) = \gamma(g^{-1}, g), \end{aligned} \tag{1.1.10}$$

for any symmetry transformation g of G. The first property is a consequence of the neutrality property of the group multiplication of G, while the second one is a consequence of the invertibility property of the group multiplication of G.

Relation (1.1.9) entails also that, for any symmetry transformation g, the operators $U(g^{-1})$ and $U(g)^{-1}$ differ by a g-dependent phase factor. Specifically, the relation between them takes the form

$$U(g^{-1}) = \varpi(g)U(g)^{-1}, \quad \varpi(g) = \gamma(g, g^{-1})^{-1},$$

In general it is not possible to absorb the phase $\gamma(g', g)$ by redefining the operators $U(g)$ into the operators $\hat{U}(g)$ given in (1.1.7) with $\alpha(g)$ a suitable phase function. By doing so $\gamma(g', g)$ gets replaced by

$$\hat{\gamma}(g, g') = \gamma(g, g')\alpha(g \cdot g')\alpha(g)^{-1}\alpha(g')^{-1}, \tag{1.1.11}$$

and in general this cannot be made equal to 1 by a judicious choice of $\alpha(g)$. A phase function $\gamma(g', g)$ of the form

$$\gamma(g, g') = \alpha(g \cdot g')\alpha(g)^{-1}\alpha(g')^{-1}, \tag{1.1.12}$$

for some function $\alpha : G \to U(1)$ is called a **2-coboundary** of G (which is a trivial 2-cocycle).

The G 2–cocycle $\hat{\gamma}(g', g)$ in (1.1.11) can be rendered equal to 1 precisely when the G 2–cocycle $\gamma(g', g)$ is a G 2–coboundary of the form (1.1.12). This may or may not happen. The study of these matters is the object of a branch of algebra known as group cohomology. Some special results will be obtained below. At any rate, if $\hat{\gamma}(g', g)$ can be made equal to 1, the representation $\hat{U}(g)$ becomes genuinely unitary, so that

$$\hat{U}(g \cdot g') = \hat{U}(g)\hat{U}(g'),$$

for all symmetry transformations g, g' of G. Then the unitary family $U(g)$ forms a genuine **unitary representation** of the symmetry group G.

1.2 | Dynamical Symmetry Groups

1.2.1 | Energy Eigenvalues Classification

1.3 | Lie Groups and Lie Algebras

A Lie group G is an infinite group that can be parametrized in a neighborhood of the neutral element by continuous parameters θ^a, $a = 1, \dots, d$, organized as a vector $\boldsymbol{\theta}$, in one–to–one fashion:

$$\begin{cases} \boldsymbol{\theta} \in D \subseteq \mathbb{R}^d, \\ g \,:\, D \mapsto G, \\ \boldsymbol{\theta} \mapsto g(\boldsymbol{\theta}), \end{cases}$$

where D is an open neighborhood of the origin in $\mathbb{R}^d$ ($\mathbf{0} \in D$ and $g(\mathbf{0}) \in G$). The number d of parameters is called the dimension of G, seen as a differential map. The group operations of multiplication and inversion are smooth functions of the parameters.

If the parametrization is smooth, as we assume, G can be thought geometrically as a group manifold. Below, we treat G as a matrix group for definiteness.

We assume that 0 belongs to the parameter space and that

$$g(\mathbf{0}) = 1,$$

which can be read as a normalizing condition. By virtue of the parametrization, for any pair of parameter vectors $\boldsymbol{\theta}$, $\boldsymbol{\theta}'$, there exists a third parameter vector $\boldsymbol{\theta}''$ such that

$$g(\boldsymbol{\theta}')g(\boldsymbol{\theta}) = g(\boldsymbol{\theta}'').$$

As the parametrization is one–to–one,[1] $\boldsymbol{\theta}''$ is also uniquely determined by $\boldsymbol{\theta}$, $\boldsymbol{\theta}'$. So, it can be expressed as a function of them

$$\boldsymbol{\theta}'' = m(\boldsymbol{\theta}', \boldsymbol{\theta}),$$

which is a function of the **group multiplication**. The function m is smooth as the group multiplication is smooth. The inverse of $g(\boldsymbol{\theta})$ is given by

$$g(\boldsymbol{\theta})^{-1} = g(\boldsymbol{\theta}'),$$

for some $\boldsymbol{\theta}' = j(\boldsymbol{\theta})$, which is unique since the parameterization is one-to-one. The function j, implementing the group inversion, is smooth as the group inversion is smooth.

We have

$$\begin{cases} m : D \times D \to D, \\ j : D \to D. \end{cases}$$

[...]

These functions have to respect some properties, which are the group multiplication properties: associativity, invertibility and neutrality. Thus given three parameter vectors $\boldsymbol{\theta}$, $\boldsymbol{\theta}'$, $\boldsymbol{\theta}''$, the following relations must hold:

$$\begin{aligned} \left(g(\boldsymbol{\theta})g(\boldsymbol{\theta}')\right) g(\boldsymbol{\theta}'') &= g(\boldsymbol{\theta}) \left(g(\boldsymbol{\theta}')g(\boldsymbol{\theta}'')\right), \\ g(m(\boldsymbol{\theta}, \boldsymbol{\theta}'))g(\boldsymbol{\theta}'') &= g(\boldsymbol{\theta})g(m(\boldsymbol{\theta}', \boldsymbol{\theta}'')), \\ g(m(m(\boldsymbol{\theta}, \boldsymbol{\theta}'), \boldsymbol{\theta}'')) &= g(m(\boldsymbol{\theta}, m(\boldsymbol{\theta}', \boldsymbol{\theta}''))), \end{aligned}$$

[1] This means that each element of the group corresponds to a unique set of parameters: $g(\boldsymbol{\theta}) = g(\boldsymbol{\theta}') \implies \boldsymbol{\theta} = \boldsymbol{\theta}'$.

since the parametrization is one–to–one, these relations [...] and so we find the three properties of the group multiplication functions:

$$\begin{aligned} m(m(\boldsymbol{\theta},\boldsymbol{\theta}'),\boldsymbol{\theta}'') &= m(\boldsymbol{\theta},m(\boldsymbol{\theta}',\boldsymbol{\theta}'')), \\ m(\boldsymbol{\theta},j(\boldsymbol{\theta})) &= m(j(\boldsymbol{\theta}),\boldsymbol{\theta}) = \mathbf{0}, \\ m(\boldsymbol{\theta},\mathbf{0}) &= m(\mathbf{0},\boldsymbol{\theta}) = \boldsymbol{\theta}. \end{aligned} \tag{1.3.1}$$

The idea now is to try and understand the structure of G by looking at the behavior of $g(\boldsymbol{\theta})$ in a neighborhood of the neutral element. This is done by Taylor expanding it in power series around $\mathbf{0}$:

$$g(\boldsymbol{\theta}) = g(\mathbf{0}) + \left.\frac{\partial g}{\partial\theta^a}\right|_{\mathbf{0}}\theta^a + \frac{1}{2}\left.\frac{\partial^2 g}{\partial\theta^a\partial\theta^b}\right|_{\mathbf{0}}\theta^a\theta^b + \mathrm{O}(\theta^3).$$

Now computing individual terms of the expansion, we have

$$g(\boldsymbol{0}) = 1, \quad t_a = \left.\frac{\partial g}{\partial\theta^a}\right|_{\mathbf{0}}, \quad t_{ab} = \left.\frac{\partial^2 g}{\partial\theta^a\partial\theta^b}\right|_{\mathbf{0}},$$

so that

$$g(\boldsymbol{\theta}) = 1 + t_a\theta^a + \frac{1}{2}t_{ab}\theta^a\theta^b + \mathrm{O}(\theta^3).$$

Let us highlight the fact that the coefficients t_a, t_{ab} are matrices, as $g(\boldsymbol{\theta})$ is a matrix group. t_{ab} in particular is symmetric in its indices a, b since the order of differentiation does not matter.

We can now think of G as an **hypersurface in the space of matrices**, containing the identity matrix. In the neighborhood of the identity, there is D, subset of our hypersurface, which is diffeomorphic to an open neighborhood of the origin in $\mathbb{R}^d$.

If we vary $g(0,\dots,0,\theta^a,0,\dots,0)$, we get a curve in D passing through the identity (when $\theta^a = 0$). The tangent vector to this curve at the identity is given by $t_a = \frac{\partial g}{\partial\theta^a}\big|_{\mathbf{0}}$. Varying all parameters θ^a we get d curves in D passing through the identity and their tangent vectors at the identity are given by t_a, $a = 1,\dots,d$. These tangent vectors are **linearly independent** since the parametrization is one–to–one. Thus, they span a vector space of dimension d.

Thus the tangent space to G at the identity is more than a vector space of dimension d (spanned by the matrices t_a, $a = 1,\dots,d$, costituing a basis). We denote this tangent space by $\mathfrak{g}$ and call it the Lie algebra of G. It is a vector space of dimension d and it is closed under the commutator of matrices. The Lie algebra $\mathfrak{g}$ encodes all the information about the local structure of G around the identity element.

We could now think to Taylor expand the group multiplication functions m, j around $\mathbf{0}$ and try to find the structure of G by looking at the properties of the coefficients of the expansion. Starting from the associativity property, we find that the coefficients of the expansion of m must satisfy

$$m^a(\boldsymbol{\theta},\boldsymbol{\theta}') = \theta^a + \theta'^a + m^a_{bc}\theta^b\theta'^c + \mathrm{O}(\boldsymbol{\theta}'^2,\boldsymbol{\theta}) + \mathrm{O}(\boldsymbol{\theta}^2,\boldsymbol{\theta}'),$$

while for the inversion property we find

$$j^a(\boldsymbol{\theta}) = -\theta^a + j^a_{bc}\theta^b\theta^c + \mathrm{O}(\boldsymbol{\theta}^3).$$

If we now recall the Taylor expansion of $g(\boldsymbol{\theta})$ around $\mathbf{0}$

$$g(\mathbf{0}) = 1 + t_a\theta^a + \frac{1}{2}t_{ab}\theta^a\theta^b + \mathrm{O}(\theta^3),$$

we can note how the coefficients t_a define another member of the Lie algebra $\mathfrak{g}$ by defining the **commutator**, or **Lie bracket**, of two basis elements t_a, t_b of $\mathfrak{g}$ as follows:

$$[t_a, t_b] = t_a t_b - t_b t_a = f_{ab}{}^c t_c \in \mathfrak{g},$$

where $f_{ab}{}^c$ are the **structure constants** of the Lie algebra $\mathfrak{g}$. The structure constants are *antisymmetric in their lower indices* a, b since the commutator is antisymmetric. These structure constants are related to the coefficients $m^a{}_{bc}$ of the expansion of the group multiplication function m by

$$f^a{}_{bc} = m^a{}_{bc} - m^a{}_{cb}.$$

If we change parametrization, the structure constants change accordingly, but the Lie algebra $\mathfrak{g}$ remains the same.

Taking again the definition of the multiplication function

$$g(\boldsymbol{\theta}')g(\boldsymbol{\theta}) = g(m(\boldsymbol{\theta}', \boldsymbol{\theta})),$$

we can Taylor expand both sides around $\mathbf{0}$ and find the following relation between the coefficients of the expansion of g and the structure constants:

$$\left[1 + \theta^a t_a + \frac{1}{2}\theta^a\theta^b t_{ab} + \mathrm{O}(\theta^3)\right]\left[1 + \theta'^a t_a + \frac{1}{2}\theta'^a\theta'^b t_{ab} + \mathrm{O}(\theta'^3)\right] \\ = [\dots]$$

and by relating the coefficients of the expansion of both sides in θ^b, θ'^a we find the following relations (renaming summation indices):

$$\begin{aligned} t_a t_b &= t_{ab} + m^c{}_{ab} t_c, \\ t_b t_a &= t_{ab} + m^c{}_{ba} t_c, \end{aligned}$$

and if we subtract the second from the first, we find

$$[t_a, t_b] = (m^c{}_{ab} - m^c{}_{ba}) t_c = f_{ab}{}^c t_c,$$

so that the structure constants $f_{ab}{}^c$ of the Lie algebra $\mathfrak{g}$ are related to the coefficients $m^c{}_{ab}$ of the expansion of the group multiplication function m by

$$f^c{}_{ab} = m^c{}_{ab} - m^c{}_{ba}.$$

Thus to sum up:

- The Lie algebra $\mathfrak{g}$ of a Lie group G is the tangent space to G at the identity element, spanned by the matrices t_a, $a = 1, \dots, d$.
- The basis elements t_a of the Lie algebra $\mathfrak{g}$ satisfy the commutation relations

  $$[t_a, t_b] = f_{ab}{}^c t_c,$$

  where $f_{ab}{}^c$ are the structure constants of the Lie algebra $\mathfrak{g}$.
- The structure constants $f_{ab}{}^c$ of the Lie algebra $\mathfrak{g}$ are related to the coefficients $m^c{}_{ab}$ of the expansion of the group multiplication function m by

  $$f^c{}_{ab} = m^c{}_{ab} - m^c{}_{ba}.$$

The Lie algebra, the tangent space to the Lie group at the identity, is not only a vector space, but also an algebra, since it is closed under the commutator of matrices. The Lie algebra encodes all the information about the local structure of the Lie group around the identity element. In particular, it encodes the structure constants ${f_{ab}}^c$ which determine the commutation relations of the basis elements t_a of the Lie algebra.

The commutator of the basis elements t_a of the Lie algebra $\mathfrak{g}$ has some properties, which are the **Jacobi identity** and the **antisymmetry** of the commutator:

$$\begin{aligned} [t_a,\, t_b] + [t_b,\, t_a] &= 0, \\ [t_a,\, [t_b,\, t_c]] + [t_b,\, [t_c,\, t_a]] + [t_c,\, [t_a,\, t_b]] &= 0. \end{aligned}$$

These two properties can be expressed in terms of the structure constants ${f_{ab}}^c$ as follows:

$$\begin{aligned} {f_{ab}}^c + {f_{ba}}^c &= 0, \\ {f_{ad}}^e {f_{bc}}^d + {f_{bd}}^e {f_{ca}}^d + {f_{cd}}^e {f_{ab}}^d &= 0, \end{aligned}$$

which are the **antisymmetry** of the structure constants in their lower indices and the **Jacobi identity** for the structure constants, respectively. This means that not every set of numbers ${f_{ab}}^c$ can be the structure constants of a Lie algebra, but only those that satisfy the antisymmetry and Jacobi identity properties.

1.3.1 | Exponential Parametrization of Lie Groups

Given the parametrization of a Lie group G in a neighborhood of the identity element we used to define the Lie algebra $\mathfrak{g}$ of G

$$g\,:\, D \mapsto G, \quad \boldsymbol{\theta} \in D \mapsto g(\boldsymbol{\theta}) \in G,$$

we can now try to find a global parametrization of G in terms of the elements of $\mathfrak{g}$. If we consider another parametrization of G in a neighborhood of the identity element defined by

$$\tilde{g}(\theta) = \lim_{N\to\infty} g\left(\frac{\theta}{N}\right)^N,$$

we obtain the **exponential map**, which provides a global parametrization of the Lie group G in terms of the elements of its Lie algebra $\mathfrak{g}$. Let us analyze its behaviour in a neighborhood of the identity element. We have

$$g\left(\frac{\theta}{N}\right)^N = \left[1 + \frac{t_a\theta^a}{N} + \frac{1}{2}\left(\frac{t_a\theta^a}{N}\right)^2 + \mathrm{O}\left(\frac{1}{N^3}\right)\right]^N \sim \left(1 + \frac{t_a\theta^a}{N}\right)^N \xrightarrow{N\to\infty} e^{t_a\theta^a},$$

which is the exponential of a matrix, defined by the power series

$$e^{t_a\theta^a} = \sum_{n=0}^{\infty} \frac{(t_a\theta^a)^n}{n!}.$$

Thus, the exponential map provides a global parametrization of the Lie group G in terms of the elements of its Lie algebra $\mathfrak{g}$ as follows:

$$\tilde{g}(\theta) = e^{t_a\theta^a} = 1 + \tilde{t}_a\theta^a + \frac{1}{2}\theta^a\theta^b\tilde{t}_{ab} + \mathrm{O}(\theta^3),$$

where

$$\tilde{t}_A = t_a, \quad \tilde{t}_{ab} = t_a t_b + t_b t_a.$$

The same goes for the he group multiplication function m and the inversion function j. We can define new functions $\tilde{m}$, $\tilde{j}$ as follows:

$$\begin{aligned} \tilde{j}(\theta) &= -\theta^a, \\ \tilde{g}(\tilde{j}(\theta)) = \tilde{g}(\theta)^{-1} = \cdots &= e^{-t_a\theta^a}, \end{aligned}$$

while for the multiplication function we have the **Campbell-Baker-Hausdorff function** so that

$$[\dots]$$

These new structure constants are not different from the original ones, since the exponential map is a reparametrization of the Lie group G in terms of the elements of its Lie algebra $\mathfrak{g}$: $\tilde{t}_a = t_a$ implies that

$$f^c{}_{ab} t_c = [t_a,\, t_b] = \left[\tilde{t}_a,\, \tilde{t}_b\right] = \tilde{f}^c{}_{ab}\tilde{t}_c = \tilde{f}^c{}_{ab} t_c,$$

thus the structure constants of the Lie algebra $\mathfrak{g}$ are the same in both parametrizations, as expected since the Lie algebra is an intrinsic property of the Lie group G and does not depend on the choice of parametrization:

$$f^c{}_{ab} = \tilde{f}^c{}_{ab}.$$

Thus we have

$$\tilde{m}^a{}_{bc} = \frac{1}{2}\tilde{f}^a{}_{bc} = \frac{1}{2} f^a{}_{bc},$$

and if we expand the CBH function in power series we find[2]

$$\tilde{m}^a(\theta,\theta') = \theta^a + \theta'^a + \frac{1}{2} f^a{}_{bc}\theta^b\theta'^c + \mathrm{O}(\theta^3,\theta'^3),$$

which is like a polynomial expansion of the group multiplication function m in terms of the structure constants $f^a{}_{bc}$ of the Lie algebra $\mathfrak{g}$. The exponential map thus provides a global parametrization of the Lie group G in terms of the elements of its Lie algebra $\mathfrak{g}$ and allows us to express the group multiplication function m in terms of the structure constants $f^a{}_{bc}$ of the Lie algebra $\mathfrak{g}$.

Example (Phase Group U(1)). Let us define the group of phase transformations $\mathrm{U}(1)$ as the group of complex numbers of unit modulus under multiplication:

$$\mathrm{U}(1) = \{z \in \mathbb{C} : |z| = 1\} = \{e^{i\theta} : \theta \in \mathbb{R}\}.$$

It is an abelian Lie group of dimension 1 which represents the Gauge group of QED. The elements of $\mathrm{U}(1)$ can be represented by an angle θ with an exponential parametrization as follows:

$$\tilde{g}(\theta) = e^{i\theta^a t_a} = e^{i\theta},$$

since the Lie algebra of $\mathrm{U}(1)$ is one-dimensional and spanned by the identity matrix $t = 1$. The structure constants of the Lie algebra of $\mathrm{U}(1)$ are zero since the commutator of any two elements of the Lie algebra is zero:

$$[t,t] = 0 \implies f_{ab}{}^c = 0.$$

[2]There exists a closed form expression for the CBH function, but it is not important for our purposes: remind only that exact complicated expressions for computing all the teerms exist, but we are only interested in the first few terms of the expansion.

Example (Special Unitary Group SU(2)). The special unitary group SU(2) is the group of 2×2 unitary matrices with determinant 1:

$$\mathrm{SU}(2) = \{U \in \mathbb{C}^{2\times 2} : U^\dagger U = \mathbb{I}_2, \det(U) = 1\}.$$

It is a non-abelian Lie group of dimension 3: $\boldsymbol{\theta} \in \mathbb{R}^3$. The elements of SU(2) can be parametrized using the Pauli matrices σ_i as follows:

$$\tilde{g}(\theta) = e^{i\theta^i \frac{\tau(\theta)}{2}},$$

where

$$\tau(\theta) = \begin{pmatrix} \theta^3 & \theta^1 - i\theta^2 \\ \theta^1 + i\theta^2 & -\theta^3 \end{pmatrix} = \theta^1 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \theta^2 \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + \theta^3 \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \theta^i \sigma_i,$$

Thus $t_a = \frac{1}{2}\tau_a$ and the structure constants of the Lie algebra $\mathfrak{su}(2)$ are given by the Levi-Civita symbol ϵ_{ijk}:

$$[\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k \implies f_{ij}{}^k = 2\epsilon_{ijk}.$$

Example (Special Orthogonal Group SO(3)). The special orthogonal group SO(3) is the group of 3×3 real orthogonal matrices with determinant 1:

$$\mathrm{SO}(3) = \{R \in \mathbb{R}^{3\times 3} : R^T R = \mathbb{I}_3, \det(R) = 1\}.$$

It is a non-abelian Lie group of dimension 3. The elements of SO(3) can be parametrized using the generators of rotations J_i as follows:

$$\tilde{g}(\theta) = e^{i\theta^i J_i},$$

where the generators J_i satisfy the commutation relations

$$[J_i, J_j] = i\epsilon_{ijk}J_k.$$

[-....-] to be finished

1.4 | Unitary Representations of Continuous Symmetries

If the symmetry group G is a Lie group, then the quantum unitary operators representing the group action on states and observables can be expressed as functions of the parameter vector $\boldsymbol{\theta}$ of G:

[...]

$$U(m(\theta,\theta')) = U(g(m(\theta,\theta'))) = U(g(\theta)g(\theta')) = \gamma(g(\theta),g(\theta'))U(g(\theta))U(g(\theta')),$$

where $\gamma(g(\theta),g(\theta'))$ is a phase factor which can be written as $\gamma(g(\theta),g(\theta')) = e^{i\omega(g(\theta),g(\theta'))}$ for some real function ω. Thus

$$U(m(\theta,\theta')) = e^{i\omega(g(\theta),g(\theta'))}U(g(\theta))U(g(\theta')).$$

Example (EXCERCISE).

??? photo

When you have a Lie Group and a projected unitary representation in the Hilbert space of states, you can always get to this relation with the phase factor γ by redefining the unitary operators $U(g)$ with a suitable phase factor, all given to the properties of the Lie group G and the properties of the projective representation U.

Since the Lie groups act homogeneously, we can always translates what happens at the identity element to any other element of the group. Thus, by looking at the properties of the group around the identity element, we can understand the properties of the whole group. By Taylor expanding $U(\boldsymbol{\theta})$ near the neutral element $\mathbf{0}$ of the group, we get

$$U(\boldsymbol{\theta}) = 1 - i\theta^a Q_a + \frac{1}{2}\theta^a\theta^b Q_{ab} + \mathrm{O}(\theta^3), \quad Q_{ab} = Q_{ba},$$

where Q_a are the generators of the representation U of the Lie group G. The generators Q_a are essentially the tangent vectors to the representation U of the Lie group G at the identity element

$$Q_a = i\frac{\partial U(\boldsymbol{\theta})}{\partial \theta^a}\Big|_{\boldsymbol{\theta}=\mathbf{0}}.$$

We are now going to prove that the generators Q_a of the representation U of the Lie group G are self-adjoint operators:

$$Q_a^\dagger = Q_a.$$

We can start from the adjoint

$$\begin{aligned} Q_a^\dagger &= \left(i\frac{\partial U(\boldsymbol{\theta})}{\partial \theta^a}\Big|_{\boldsymbol{\theta}=\mathbf{0}}\right)^\dagger \\ &= -i\frac{\partial U^\dagger(\boldsymbol{\theta})}{\partial \theta^a}\Big|_{\boldsymbol{\theta}=\mathbf{0}} \\ &= -i\frac{\partial U^{-1}(\boldsymbol{\theta})}{\partial \theta^a}\Big|_{\boldsymbol{\theta}=\mathbf{0}} \\ &= +iU(\boldsymbol{\theta})^{-1}\frac{\partial U(\boldsymbol{\theta})}{\partial \theta^a}U(\boldsymbol{\theta})^{-1}\Big|_{\boldsymbol{\theta}=\mathbf{0}} = Q_a, \end{aligned}$$

where we have used the unitarity of U and the fact that $U(\mathbf{0}) = 1$. [comments]

We have another property to prove: we have to normalize the phase function ω

$$\omega(\boldsymbol{\theta},\mathbf{0}) = \omega(\mathbf{0},\boldsymbol{\theta}) = 0,$$

so that we remember that

$$\omega(\boldsymbol{\theta}, \boldsymbol{\theta}') = \theta^a \theta'^b \omega_{ab} + \mathrm{O}(\theta^3),$$

so that the phase factor γ is normalized to 1 at the identity element of the group, since

$$e^{i\omega(\mathbf{0},\boldsymbol{\theta})} = \gamma(g(\mathbf{0}), g(\boldsymbol{\theta}')) = 1 = e^{i\omega(\boldsymbol{\theta},\mathbf{0})} = \gamma(g(\boldsymbol{\theta}), g(\mathbf{0})) = 1.$$

???

[...]

$$[Q_a, Q_b]$$

... as we did for $g(\theta)g(\theta')$ we can multiply the taylor expansion of $U(\theta)$ and $U(\theta')$ and compare the result with the taylor expansion of $U(m(\theta, \theta'))$ to get

$$U(\boldsymbol{\theta})U(\boldsymbol{\theta'}) = e^{-i\omega(\boldsymbol{\theta},\boldsymbol{\theta'})} U(m(\boldsymbol{\theta}, \boldsymbol{\theta'})).$$

If we insert the taylor exapnsions of

- $U(\boldsymbol{\theta}) = 1 - i\theta^a Q_a + \frac{1}{2}\theta^a\theta^b Q_{ab} + \mathrm{O}(\theta^3)$
- $\omega(\boldsymbol{\theta}, \boldsymbol{\theta'}) = \theta^a \theta'^b \omega_{ab} + \mathrm{O}(\theta^3)$
- $m(\boldsymbol{\theta}, \boldsymbol{\theta'}) = \boldsymbol{\theta} + \boldsymbol{\theta'} + \frac{1}{2} f^c_{ab}\theta^a\theta'^b + \mathrm{O}(\theta^3)$

we get a mess, but if we compute everything and compare the coefficients of the terms of order $\theta^a\theta'^b$ we get

$$Q_a Q_b = i m^c_{ab} Q_c + i\omega_{ab}\mathbb{I} - Q_{ab},$$

and since we wanted to compute the commutator $[Q_a, Q_b]$ we can compute

$$Q_b Q_a = i m^c_{ba} Q_c + i\omega_{ba}\mathbb{I} - Q_{ba},$$

so that subtracting the two we get

$$[Q_a, Q_b] = i(m^c_{ab} - m^c_{ba})Q_c + i(\omega_{ab} - \omega_{ba})\mathbb{I} = i f^c_{ab} Q_c + i\kappa_{ab}\mathbb{I},$$

where we have recalled the structure constants of the Lie algebra $\mathfrak{g}$ of the Lie group G and we have defined $\kappa_{ab} = \omega_{ab} - \omega_{ba}$. This last coefficients κ_{ab} are called the **central charges** of the representation U of the Lie group G. A central algebra is an algebra where the central charges are zero, so that the commutator of the generators Q_a of the representation U of the Lie group G is just

$$[Q_a, Q_b] = i f^c_{ab} Q_c.$$

The properties of the commutators of the generators Q_a of the representation U can be expressed as

- $[Q_a, Q_b] = i f^c_{ab} Q_c + i\kappa_{ab}\mathbb{I}$
- $\kappa_{ab} = \omega_{ab} - \omega_{ba}$
- $[Q_a, Q_b] + [Q_b, Q_a] = 0 \implies \kappa_{ab} + \kappa_{ba} = 0$
- $[Q_a, [Q_b, Q_c]] + [Q_b, [Q_c, Q_a]] + [Q_c, [Q_a, Q_b]] = 0$ (Jacobi identity)

The structure constants respects those identity in the following form

- $f^a_{bc} + f^a_{cb} = 0$
- $f^d_{ab} f^e_{dc} + f^d_{bc} f^e_{da} + f^d_{ca} f^e_{db} = 0$ (Jacobi identity)

and along with the central charges they respect the following identity

$$\kappa_{ad} f^d_{bc} + \kappa_{bd} f^d_{ca} + \kappa_{cd} f^d_{ab} = 0,$$

which is a consequence of the Jacobi identity for the generators Q_a of the representation U of the Lie group G. A set of coefficients f^c_{ab} and κ_{ab} which respect the above identities is called a **Lie algebra $\mathfrak{g}$ 2-cocycle**. This naming is consistent since the central charges κ_{ab} are related to the phase factor γ of the projective representation U of the Lie group G by the relation $\kappa_{ab} = \omega_{ab} - \omega_{ba}$.

...

$$U(g, g') = \gamma(g, g') U(g) U(g'),$$

$$U(g) \to \hat{U}(g) = \alpha(g) U(g),$$

$$\gamma(g, g') = \alpha(g, g') \alpha(g)^{-1} \alpha(g')^{-1},$$

so that we can redefine the unitary operators $U(g)$ of the projective representation U of the Lie group G by a suitable phase factor $\alpha(g)$ to get a new projective representation $\hat{U}$ of the Lie group G with a new phase factor $\hat{\gamma}$ which is related to the old one by the relation

$$\hat{\gamma}(g, g') = \alpha(g, g') \alpha(g)^{-1} \alpha(g')^{-1} \gamma(g, g'),$$

so that if we can find a suitable phase factor $\alpha(g)$ such that $\hat{\gamma}(g, g') = 1$ for all $g, g' \in G$, then we can redefine the unitary operators $U(g)$ of the projective representation U of the Lie group G by a suitable phase factor $\alpha(g)$ to get a new projective representation $\hat{U}$ of the Lie group G with a trivial phase factor $\hat{\gamma}$, so that the new projective representation $\hat{U}$ of the Lie group G is actually a linear representation of the Lie group G. However, in general, it is not always possible to find such a phase factor $\alpha(g)$, so that there are some projective representations of the Lie group G which cannot be redefined to get a linear representation of the Lie group G.

???

$$Q_a \to \hat{Q}_a = Q_a + k_a \mathbb{I},$$

where k_a are the central shifts of the generators Q_a of the representation U of the Lie group G. These shifts are related to the central charges κ_{ab} by the relation

$$\kappa_{ab} = f^c_{ab} k_c,$$

which defines a **2-coboundary** Lie algebra. A 2-coboundary Lie algebra is special case of a 2-cocycle Lie algebra, where the central charges κ_{ab} can be expressed as a linear combination of the structure constants f^c_{ab} with coefficients given by the central shifts k_a. This is **Lie Algebra Cohomografy**

1.4.1 | Exponential Parametrization

Starting from the exponential parametrization of the Lie group G

$$\tilde{g}(\boldsymbol{\theta}) = \lim_{n\to\infty} g(\frac{\theta}{n})^n = e^{\theta^a T_a},$$

while for the inverse and the multiplication we have

$$\begin{cases} \tilde{j}(\boldsymbol{\theta}) = -\theta^a, \\ \tilde{m}(\boldsymbol{\theta}, \boldsymbol{\theta}') = \boldsymbol{\theta} + \boldsymbol{\theta}' + \frac{1}{2} f^c_{ab}\theta^a\theta'^b + \mathrm{O}(\theta^3), \end{cases}$$

where the last function is the Baker-Campbell-Hausdorff formula (BCH) for the multiplication of the Lie group G in the exponential parametrization.

We can now use this exponential parametrization of the Lie group G to get an exponential parametrization of the projective representation U of the Lie group G:

$$\tilde{U}(g(\boldsymbol{\theta})) = U(\tilde{g}(\boldsymbol{\theta})) = U(e^{\theta^a T_a}) = 1 - i\theta^a Q_a + \frac{1}{2}\theta^a\theta^b Q_{ab} + \mathrm{O}(\theta^3).$$

For the phase function ω we have

$$\tilde{\omega}(\boldsymbol{\theta}, \boldsymbol{\theta}') = \tilde{\omega}_{ab}\theta^a\theta'^b + \mathrm{O}(\theta^3).$$

At the linear level there are no differences between g and $\tilde{g}$, they only manifest from the second order in the parameters θ^a. Thus we can derive

$$Q_a = \tilde{Q}_a,$$

and for the phase function ω we have

$$\tilde{Q}_{ab} = \frac{1}{2}((\omega_{ab} + \omega_{ba})\mathbb{I} - Q_a Q_b - Q_b Q_a).$$

Now we are interested in knowing if the following is true:

$$\tilde{U}(\theta) = \exp(i\theta^a Q_a),$$

which is not the case: this is a projected representation, so this will be proven to be true only up to a phase factor. We can start from the exponential parametrization of the projective representation U of the Lie group G

$$\tilde{U}(\boldsymbol{\theta}) = U(\tilde{g}(\boldsymbol{\theta})) = \lim_{n\to\infty} U(g(\frac{\theta}{n}))^n = \lim_{n\to\infty} U((\frac{1}{n^2})^n) \neq \lim_{n\to\infty} U((\frac{1}{n^2}))^n,$$

where we highlighted the difference between this projected and a true representation. Computing further we get

$$\tilde{U}(\boldsymbol{\theta}) = \exp\left(i\sum_{k}^{n}\omega(\frac{\theta}{n}, \frac{k\theta}{n})\right)U(\frac{\boldsymbol{\theta}}{n})^n,$$

where the last term, as we take the limit $n \to \infty$, becomes

$$\lim_{n\to\infty} U(\frac{\boldsymbol{\theta}}{n})^n = (1 - i\frac{\theta^a}{n}Q_a)^n = e^{-i\theta^a Q_a},$$

so that the exponential parametrization of the projective representation U of the Lie group G is given by

$$\tilde{U}(\boldsymbol{\theta}) = e^{i\alpha(\boldsymbol{\theta})}e^{-i\theta^a Q_a}.$$

If we now want to get .....

$$e^{i\alpha(\boldsymbol{\theta})-i\theta^a k_a}\tilde{U}(\boldsymbol{\theta}) = e^{-i\theta^a \hat{Q}_a},$$

where $\hat{Q}_a = Q_a + k_a\mathbb{I}$ are the shifted generators of the representation U of the Lie group G: these operates in the same way since they differ only by a phase factor, but they have different central charges.

Now we found a true representation

- $\hat{\tilde{U}}(\tilde{j}(\boldsymbol{\theta})) = \hat{\tilde{U}}(\boldsymbol{\theta})^{-1}$
- $\hat{\tilde{U}}(\tilde{m}(\boldsymbol{\theta}, \boldsymbol{\theta'})) = \hat{\tilde{U}}(\boldsymbol{\theta})\hat{\tilde{U}}(\boldsymbol{\theta'})$

The term $\exp(-i\theta^a k_a)$ is an important **phase factor**, since it is the one which allows us to get a true representation from a projective representation by shifting the generators Q_a of the representation U of the Lie group G by a suitable central shift k_a. This is the reason why we introduced the concept of 2-coboundary Lie algebra, since it allows us to understand when a projective representation of a Lie group can be redefined to get a true representation of the Lie group.

This is a **multi-valued function** on the underlying group manifold G, since the phase factor usually is not an integer which comes back to the original value after a loop around the group manifold G (it is not guaranteed that the passage is single-valued): the exponential parametrization of the projective representation U of the Lie group G is a multi-valued function on the underlying group manifold G. Usually then we introduce a branch cut in the complex plane to make it single-valued, in order to treat multi valued function as single valued function on the more complex Riemann surface (such as for the square root and the logarithm).

In our case the exponential parametrization of the projective representation U of the Lie group G is a multi-valued function on the underlying group manifold G, but can be made single-valued on a infinite Riemann surface which is a covering space of the underlying group manifold G: a covering group $\tilde{G}$. It is the same idea of the Riemann surface, but with more subtleties and complications, since the group structure has to be preserved.

There can be made analogies with the various covering groups we can find in physics, such as the universal covering group of the Lorentz group, which is the Spin group, and the universal covering group of the rotation group $SO(3)$, which is the special unitary group $SU(2)$. The projective representations of the Lorentz group and the rotation group $SO(3)$ can be redefined to get true representations of their respective covering groups, which are the Spin group and the special unitary group $SU(2)$. This is a consequence of the fact that the central charges of the projective representations of the Lorentz group and the rotation group $SO(3)$ can be expressed as linear combinations of the structure constants of their respective Lie algebras with coefficients given by suitable central shifts

$$\mathrm{SL}(2,\mathbb{C}) = \widetilde{\mathrm{SO}^+(1,3)}, \quad \mathrm{SU}(2) = \widetilde{\mathrm{SO}(3)}.$$

Appendices

A | Notation and Conventions

B | Computations and further developments